Rajalakshmi Engineering College

Department of Computer Science and Design

GE23627 DESIGN THINKING AND INNOVATION

AY 2026-2027

S.C.O.P.E.S – Team Scope Sheet

Scoping of design challenge When you want to work on real project using Design Thinking, begin the process with a specific and purposeful situation or problem to solve called Design Challenge. The Design Challenge should be doable, clear, actionable and hopefully inspiring. Your problem statement should be an outcome and not a solution.
You can use SCOPES as a tool to brainstorm and frame your problem.

TEAM NAME	**GAURDIAN**
TEAM MEMBERS	**PRADEEPKUMAR R** **241701042** **MAAHESH S** **241701025**
DOMAIN NAME	**Parent-Verified Hostel Access Management System**
DATE	
FACULTY INCHARGE	**Dr. P Revathy**

S – Situation / Problem	**S – Situation / Problem**
• What is the real situation you want to address? • Where does this problem occur? • Describe without mentioning any solution.	• In many college hostels, students need permission to leave the hostel for personal, academic, medical, or emergency reasons.

	• Gate-pass approval and student movement are often managed through manual forms, registers, or basic digital systems. • Parents may not know when their child has actually left the hostel, where the child is travelling, or whether the child has returned safely. • There can be a gap between **parental awareness, warden approval, and security-gate records**. • During travel, if a student faces an unexpected problem, there may be no quick and structured way to inform the parent or hostel authorities. • These problems mainly occur in **college and university hostel environments**, particularly during student outings and travel outside the campus. • The situation becomes more concerning when students leave without proper parental awareness, return late, or face safety issues while travelling.
C – Constraints What are the CONSTRAINTS when trying to address the situation or problem?	**C – Constraints** The major constraints when addressing this hostel student safety and gate-pass problem are: • **Privacy:** Students' personal information and location data must be protected and should not be unnecessarily exposed. • **Parent Verification:** The system must ensure that approval genuinely comes from the registered parent and not from the student. • **Internet Connectivity:** Students, parents, wardens, and security staff may not always have a stable internet connection. • **Mobile Network:** The automatic parent call and emergency communication depend on mobile network availability.

	• **GPS Accuracy:** Student location may not always be accurate, especially inside buildings, crowded areas, or places with weak GPS signals. • **Emergency Response:** Critical emergencies require reliable escalation to appropriate emergency services; the system cannot guarantee that authorities will respond immediately. • **False Emergency Alerts:** Students may accidentally press the SOS button or misuse it, creating unnecessary alerts. • **Battery Usage:** Continuous GPS/location tracking can consume significant battery power on the student's phone. • **Data Security:** Gate-pass records, parent details, phone numbers, QR codes, and location information require strong security. • **System Availability:** The system should continue to provide essential gate verification even during temporary server or network failures. • **Cost:** SMS, automated voice calls, maps/GPS services, cloud hosting, and other third-party services may involve operating costs. • **User Adoption:** Students, parents, wardens, and security guards must be willing and able to use the system correctly. • **Administrative Control:** The system must respect hostel rules and ensure that technology does not completely replace the authority of wardens and administrators. • **Scalability:** The system should work for a small hostel as well as a large college with thousands of students. • **Legal and Ethical Requirements:** Location tracking, emergency alerts, and sharing student information must follow applicable privacy, safety, and data-protection requirements.

O – Objectives & Outcomes What are the OBJECTIVES and intended OUTCOME of this design challenge?	**Objectives** • To make the **hostel gate-pass process more secure, transparent, and organized**. • To ensure that **parents are aware of and can verify their child's gate-pass requests**. • To prevent students from **misusing or bypassing parental approval**. • To maintain accurate records of **student exit and entry times**. • To allow parents to **monitor their child's location during an approved outing**, while respecting privacy. • To provide students with a **quick emergency/SOS mechanism** when they face problems outside the hostel. • To ensure that appropriate people, such as **parents, wardens, and emergency services**, can be alerted when necessary. • To reduce manual paperwork and make hostel administration more efficient. • To provide wardens and administrators with a **centralized record of gate passes, movements, late returns, and emergencies**. **Intended Outcomes** • **Safer students:** Students have a reliable way to request help during emergencies. • **Better parental awareness:** Parents know when their child leaves, where they are during an approved trip, and when they return. • **Stronger accountability:** Every gate-pass approval, QR scan, exit, and entry is recorded with a time. • **Reduced misuse:** Parent verification makes it harder for students to create unauthorized gate passes.

	• **Faster emergency communication:** Safety alerts can reach the appropriate contacts quickly. • **Improved hostel management:** Wardens and administrators can monitor and manage student movements efficiently. • **Less paperwork:** Digital records replace manual gate-pass registers. • **Better transparency:** Students, parents, wardens, and security staff have clearly defined roles and access to relevant information. **Main Intended Outcome** **To create a secure and transparent hostel environment where student movement is properly authorized, recorded, and communicated to parents and hostel authorities, while providing students with quick access to safety assistance during emergencies.**
P – People Who are the PEOPLE that you are trying to help? Whose problem are you trying to address? Primary Stakeholders Secondary Stakeholders Decision Makers	**P – People** **Primary Stakeholders** • **Hostel Students** – Need a safe and reliable way to manage their outings and emergencies. • **Parents/Guardians** – Need awareness of their child's movement and safety. • **Wardens** – Need to monitor and manage student movements. **Secondary Stakeholders** • **Security Guards** – Manage student entry and exit at the hostel gate. • **Hostel Administration Staff** – Maintain student and hostel records. • **College Management** – Monitor overall hostel safety and administration. **Decision Makers**

	• **Hostel Warden** • **Hostel/College Administrator** • **College Management** • **Principal / Dean**
E – Estimates What are the ESTIMATED resources required for this design challenge?	**E – Estimates** • **Technology:** Mobile/web app, database, GPS/maps, QR code, and notification system. • **Hardware:** Smartphones for students, parents, and security staff; computer for warden/admin. • **People:** App developers, UI/UX designer, database developer, and hostel staff for testing. • **Services:** Internet, cloud server, SMS/voice-call service, and map/GPS service. • **Time:** Approximately **2–4 months** for developing and testing a basic working prototype. • **Cost:** Mainly for **cloud hosting, voice calls, SMS, and map services**; a college prototype can be developed with low initial cost using free/developer plans.
S – Scope to Explore Is there ample SCOPE to explore multiple alternatives to address the problem?	There is **wide scope for exploring different alternatives**. • **Digital Gate Pass** – Mobile-based application instead of paper forms. • **Parent Approval** – Parent app approval, OTP, or automated voice-call approval. • **QR Verification** – QR-based student entry and exit. • **GPS Tracking** – Location sharing during approved outings. • **Emergency SOS** – Quick emergency alerts with location.

	• **Notifications** – Push notifications, SMS, email, and voice calls. • **Late Return Detection** – Automatically identify students who return late. • **AI Assistance** – Identify unusual gate-pass patterns and help wardens prioritize requests. • **Geofencing** – Alert when students move outside approved areas, if appropriate and consented. • **Digital ID** – Integrate gate pass with student ID cards. • **Future Security Features** – Face verification and integration with campus security systems.
Frame the Design Challenge	**Main Design Challenge** **How might we design a secure and transparent hostel management system that allows students to request gate passes safely, enables parents to verify and stay informed about their child's movement, helps wardens and security staff monitor entry and exit, and provides quick assistance during emergencies?** **Short Version**

	How might we make hostel students' travel safer while keeping parents and hostel authorities informed and connected? **Key Focus Areas** **Student Safety + Parent Verification + Location Awareness + QR Security + Emergency Support + Hostel Management**.

SIGNATURE